MIL-E-1/175H
12 January 1979
SUPERSEDING
MIL-E-1/175G
20 April 1971
(See Note 7)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5902

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, subminiature, beam power

Outline	---	3-3 (EIA)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	g1	k, g3	h	k, g3	a	h	g2	k, g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rk	Rg1	Ik	Pp	Pg2	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	v	Ohms	Meg	mAdc	W	W	°C	ft
Maximum:	6.6	165	0, -55	155	200	---	0.55	50	3.7	0.4	220	Note 1
Minimum:	6.0	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	110	0	110	0	270	---	---	---	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1256	Electrode current (1) (anode)	See note 2	0.4	II	Ib	23	37	mAdc
1256	Electrode current (2) (anode)	Ec1 = -40 Vdc; Rk = 0	0.4	II	Ib	---	100	µAdc
1266	Total grid current (1)	Rg1 = 1.0 Meg (see note 2)	0.4	II	Ic1	0	-1.0	µAdc
1301	Heater current		0.4	II	If	420	480	mA
1336	Heater-cathode leakage		0.4	II	Ihk	---	15	µAdc
1341	Power output (1)	Esig = 6.4 Vac; Rp = 3,000 ohms	0.4	II	Po	0.75	---	W
1201	Short and discontinuity dection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		2.5	I	R	50	---	Meg
1031	Low-frequency vibration	Rp = 2,000 ohms; 15 G; F = 40 Hz	2.5	I	Ep	---	100	mVac
1246	Audio frequency noise	Ecal = 150 mVac; Ecc2 = 110 Vdc; Ec1 = -8.7 Vdc; Rk = 0; Rp = 2,000 ohms; Rg1 = 0.5 Meg; Rg2 - 10,000 ohms; Cg2 = 4.0 µF	2.5	I	EB	---	17	vu
1256	Electrode current (screen)		2.5	I	Ic2	0	4.0	mAdc
1266	Grid emission	Ef = 7.5 V; Ec1 = -40 Vdc; Rg1 = 1.0 Meg; Rk = 0 (see note 3)	2.5	I	Is(g1)	0	-2.0	µAdc
1306	Transconductance		2.5	I	Sm	3,500	4,900	µmhos
1311	Anode resistance		---	---	rp	0.01	---	Meg
1331	Direct-interelectrode capacitance	0.405 in. dia shield	6.5	Code F	Cpg Cin Cout	--- 5.5 6.5	0.20 7.5 8.5	pF pF pF
1341	Power output (2)	Ef = 5.7 V; Esig = 6.4 Vac; Rp = 3,000 ohms	2.5	I	ΔPo Ef	---	15	%
1116	Lead fatigue		2.5	Code F	---	---	---	---
2126	Envelope strain		6.5	I	---	---	---	---
1041	Shock	450 G; Ehk = +100 Vdc; Rg1 = 0.1 Meg	---	---	---	---	---	---
1031	Vibration fatigue test	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	See note 4	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 (see note 2) -Continued							
---	Post-shock and vibration-fatigue test end points:							
1031	Low-frequency vibration		---	---	Ep	---	300	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	40	µAdc
1341	Change in power output (1) of individual tubes		---	---	ΔPo t	---	20	%
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1 ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1516	Stability life test	Eb = Ec2 = 100 Vdc; Ehk = +200 Vdc; Rg1 = 0.47 Meg; Rk = 220 ohms; TA = room	---	---	---	---	---	---	---	---
---	Stability life test end point:									
1341	Change in power output (1) of individual tubes		---	---	---	---	ΔPo t	---	10	%
1506	Heater-cycling life test	Ef = 7.0 V; 1 min "on"; 4 min "off"; Ehk = 140 Vac; Ec1 = Ec2 = Eb = 0	2.5	Code H	---	---	---	---	---	---
---	Heater-cycling life - test end point:									
	Heater-cathode leakage		---	---	---	---	Ihk	---	40	µAdc
1501	Intermittent life test (room temperature)	Stability life-test, or equivalent conditions; TA = room (see note 5)	---	---	---	---	---	---	---	---
---	Intermittent life-test end point: (room temperature (500 hours)									
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1 ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
---	Intermittent life-test end point: (room temperature)(1,000 hours)									
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
1501	Intermittent life test	Stability life-test conditions; TE = +220°C (min) (see note 6)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points: (500 hours)									
---	Inoperatives		---	---	1	3	---	---	---	---
1301	Heater current		----	---	1	3	If	414	492	mA
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	60	µAdc
1266	Total grid current (1)		---	---	1	3	Icl	0	-2.0	µAdc
1341	Change in power output (1) of individual tubes		---	---	1	3	ΔPo t	---	20	%
1341	Power output (1), average change		---	---	---	---	Avg ΔPo t	---	15	%
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
1341	Power output (2)		---	---	1	3	ΔPo Ef	---	15	%
---	Total defectives		---	---	3	6	---	---	---	---
---	Intermittent life-test end points: (1,000 hours)									
---	Inoperatives		---	---	1	3	---	---	---	---
1301	Heater current		---	---	1	3	If	414	492	mA
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	60	µAdc
1266	Total grid current (1)		---	---	1	3	Icl	0	-2.0	µAdc
1341	Change in power output (1) of individual tubes		---	---	1	3	ΔPo t	---	25	%
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
1341	Power output (2)		---	---	1	3	ΔPo Ef	---	20	%
---	Total defectives		---	---	4	8	---	---	---	---

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

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NOTES: -Continued.

2. This test to be performed at the conclusion of the holding period.

3. Prior to this test, the tube shall be preheated a minimum of 5 minutes at the conditions specified below. The 3-minute test is not permitted. Test at preheat conditions within 3 seconds after preheating. Total grid current (2) shall be the last test performed on the sample selected for the total grid current (2) test.

	Ef	Ec1	Ec2	Ec3	Eb	Rk	Rg1
	V	Vdc	Vdc	Vdc	Vdc	Ohms	Meg
	7.5	0	100	0	100	220	0.47

4. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

5. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

6. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey IB (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

7. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 99
DLA - ES

User activities:
Army -
Navy - AS, CG, MC, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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